

Knowledge Transfer

1. Data Mover
2. Database Setup
3. Database Configuration Wizard
4. Source Code Walkthroughs

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Introduction

- Purpose of knowledge transfer
 - Introduction of DataMover
 - Some background on existing features
 - Runtime and source information
- Scope of discussion
 - General understanding of DataMover
 - Database Configuration Wizard Integration
 - Data Mover commands
 - Source Code Walkthroughs

Topics of Discussion

Session 1:

- Data Mover
- Database Configuration Wizard (DCW)
- Data Mover Commands

Session 2:

- Data Mover Source Code
- Source Code Walkthroughs
- Q&A

Session 1

Understanding the PeopleSoft DataMover

- Transfer application data between PeopleSoft databases
- Move PeopleSoft databases across operating systems and database platforms
- Execute Data Mover script (dms)
- Execute Structured Query Language (SQL) statements against any PeopleSoft database, regardless of the underlying operating system or database platform
- Create, edit, and run scripts
 - These scripts may include any combination of SQL commands and PeopleSoft Data Mover commands for exporting and importing data.
- Database Setup utility

Data Mover: Supported file types

1. Data Mover Script files (.dms)

These are the files created using PeopleSoft Data Mover

2. Query files (.sql)

These are the files created using the Build SQL functionality in PeopleSoft Application Designer or using a query tool specific to a relational database management system (RDBMS), such as PL/SQL on Oracle

Understanding the PeopleSoft DCW

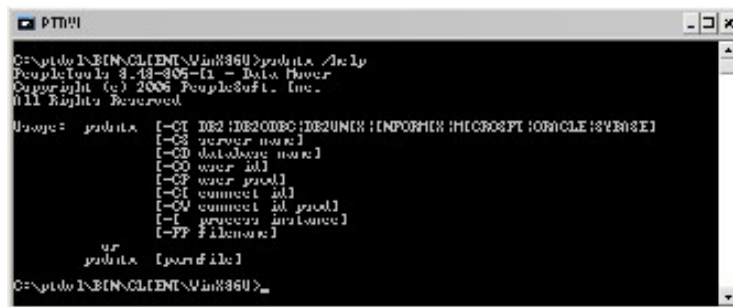
The Database Configuration Wizard (DCW) is a tool designed to simplify the PeopleSoft database installation process. When you run DCW, Data Mover is running silently in the background process—run script and import data.

Using DataMover

1. Data Mover development environment (psdmt.exe)



2. Data Mover command-line interface (psdmtx)



DataMover Command Line

- The PeopleSoft Data Mover command line program is located as follows:
 - On Microsoft Windows systems — %PS_HOME%\bin\client\winx86
 - On UNIX systems — \$PS_HOME/bin
- Standard Command Line Syntax
psdmtx -CT *dbtype* [-CS *server*] -CD *database_name* -CO *user_ID* -
CP *user_password* [-CI *connect_ID* -CW *connect_password*]
[-I *process_instance*] -FP *dms_filepath*
- Command Line Help
psdmtx /help

DataMover Operating Modes

- Regular mode

Most of the time, you use regular mode. To sign in to regular mode, enter your **PeopleSoft user ID** and password during sign-in. In regular mode, all commands are valid.

- Bootstrap mode

In bootstrap mode, you use a **database access ID** and password when signing in.

Configuration

Windows:

```
%PS_HOME%\bin\client\winx86\pscfg.exe
```

UNIX:

```
cd pt848
```

```
. ./psconfig.sh
```

Environment Variables

- Windows:

%PS_HOME%\data

%PS_HOME%\scripts

%PS_HOME%\log

- UNIX:

\$PS_DM_DATA (\$PS_HOME/data)

\$PS_DM_SCRIPT (\$PS_HOME/scripts)

\$PS_DM_LOG (\$PS_HOME/log)

Setup SQL Trace

- Windows:
 - Run Configuration Manager (pscfg.exe)
 - From the trace tab: check the trace option
- Unix:
 - `export PS_SERVER_CFG=$PS_HOME/setup/psdmtx.cfg`
 - `export PS_SERVDIR=$PS_HOME/log`
 - To configure tracing for PeopleSoft Data Mover, you must edit the psdmtx.cfg file to specify the appropriate tracing behavior

Setup SQL Trace on UNIX

```
export PS_SERVER_CFG=$PS_HOME/setup/psdmtx.cfg
```

```
[Trace]
```

```
=====
; Server Trace settings
=====
;
; SQL Tracing Bitfield
;
; Bit      Type of tracing
; ---      -
; 1        - SQL statements
; 2        - SQL statement variables
; 4        - SQL connect, disconnect, commit and rollback
; 8        - Row Fetch (indicates that it occurred, not data)
; 16       - All other API calls except ssb
; 32       - Set Select Buffers (identifies the attributes of columns
;           to be selected).
; 64       - Database API specific calls
; 128      - COBOL statement timings
; 256      - Sybase Bind information
; 512      - Sybase Fetch information
; 4096     - Manager information
; 8192     - Mapcore information
; Dynamic change allowed for TraceSql and TraceSqlMask
```

```
TraceSql=63
```

```
TraceSqlMask=12319
```

Using the Database Setup utility

To access the Database Setup utility:

- Sign in to PeopleSoft Data Mover in bootstrap mode
- Use the access ID and password rather than your PeopleSoft user ID and password
- Select File, Database Setup

Database Setup – 8.48

- Remove the disable trigger option from “Database Setup-Select People Application”
- Generate the import script for PTools 8.48 and App 9.0 or newer.

Database Setup – dbcode.xx

[LicenseInfo]

LICENSE_GROUP = 06

LICENSE_CODE = ffffffffffffffffffffffffffff7

PRODUCT_VERSION=9.0

[dbcodes]

TOOLS = PeopleTools,,

PT = PeopleTools System Database,,

PTLP = PeopleTools Language Pack,,

PTLDK = PeopleTools Language Development Kit,,

Understanding Data Mover Import Scripts

- When you run the script, Data Mover typically does the following:
 - `IMPORT *`
<Create all the PeopleTools and application>
 - `ENCRYPT_PASSWORD *`
<Encrypt security information for the database>
 - `CREATE_TRIGGER *`
<Create application required triggers>
 - `REPLACE_VIEW *`
<Create PeopleSoft views>
 - `CREATE_TEMP_TABLE *`
<Create PeopleSoft temporary tables>

Running the Database Configuration Wizard

When you run DCW, Data Mover does the following:

1. `IMPORT *` Create all the PeopleTools and application tables with their indexes.
2. `ENCRYPT_PASSWORD *` Encrypt security information for the database.
3. `CREATE_TRIGGER *` Create application required triggers.
4. `REPLACE_VIEW *` Create PeopleSoft views.
5. `CREATE_TEMP_TABLE *` Create PeopleSoft temporary tables.

Running the Database Configuration Wizard

1. Windows:

- Go to `<PS_HOME>\setup\DatabaseInstall`
setup.exe

2. Unix:

- Go to `<PS_HOME>/setup/databaseinstall`
- Launch the installation using the following command:
setup.xxx -is:javaconsole -console

Note: where xxx is the operating system platform.

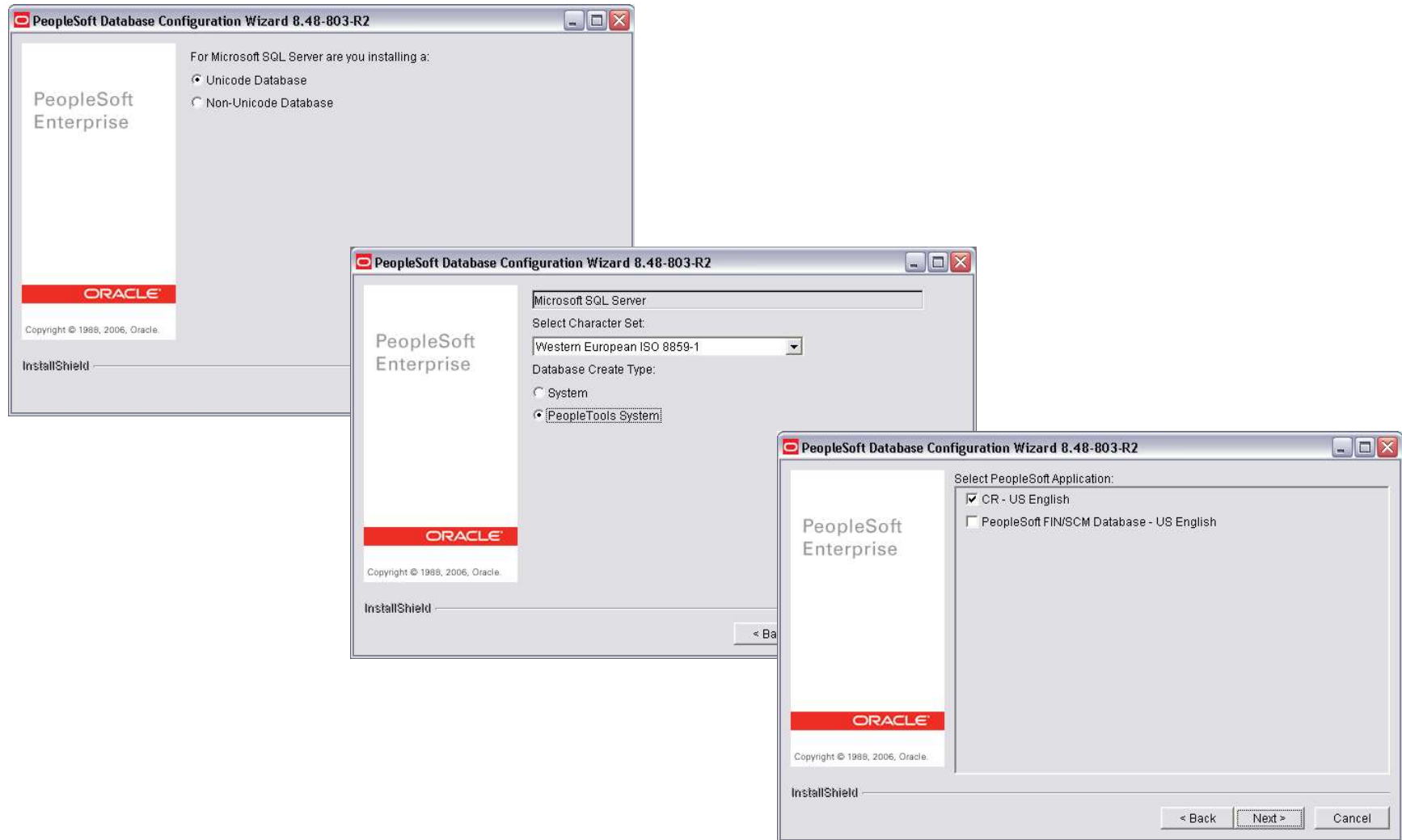
DataMover – Database Setup

The image displays three overlapping windows from the DataMover Database Setup wizard:

- Database Setup**: The main configuration window. It shows "Select Target Database" as "Microsoft SQL Server". Under "Database Type", "Non-Unicode" is selected. "Select Character Set" is set to "Western European ISO 8859-1".
- Database Setup - Select PeopleSoft Application**: A sub-window showing available applications. "CR - US English" and "PeopleSoft FIN/SCM Database - US English" are listed. Under "Data Mover Scripts to Create", "PeopleTools System Database - US English" is selected. At the bottom, "System" is selected for "Database Type".
- Database Setup - Database Parameters**: A sub-window for database-specific parameters. The fields are: "Database Name" (PTSYS), "Symbolic ID" (sa1), "Access ID" (sa), "Access Password" (masked with asterisks), and "Connect ID" (people). Navigation buttons at the bottom include "< Back", "Finish", and "Cancel".

Buttons at the bottom of the "Database Setup" window include "< Back", "Next >", and "Cancel".

DCW – Database Setup



DCW – dbsetup.xml

XML file contains database setup information.

Example:

```
<?xml version="1.0"?>
<!-- Created by ISMP - DBSETUP -->
<!-- Current Date/Time is Thu April 10 12:05:25 PST 2006 -->

<DBSETUP>
  <DBINFO>
    <DBNAME>XXXDBNAMEXXX</DBNAME>
    <DATETIME> Thu April 10 12:05:25 PST 2006</DATETIME>
    <LICENSE_GROUP>06</LICENSE_GROUP>
    <LICENSE_CODE>ffffffffffffffffffff2</LICENSE_CODE>
    <BASE_LANGUAGE>ENG</BASE_LANGUAGE>
    <CREATE_TYPE>2</CREATE_TYPE>
    <LOG_DIR>$PS_DM_LOG</LOG_DIR>
    <SCRIPT_DIR>$PS_DM_SCRIPT</SCRIPT_DIR>
    <INPUT_DIR>$PS_DM_DATA</INPUT_DIR>
    <OUTPUT_DMS_FILE>ptsys.dms</OUTPUT_DMS_FILE>
  </DBINFO>
  <CHARSET>
    ...
  </CHARSET>
  <PARAMETER>
    ...
  </PARAMETER>
  <PSAPPS>
    ...
  </PSAPPS>
</DBSETUP>
```

Command Export

Example:

```
SET LOG $PS_HOME/log/ptengs.log;  
SET OUTPUT $PS_HOME/data/ptengs.db;  
EXPORT *;
```


Command Import

Example:

```
SET LOG $PS_HOME/log/ptengs.log;  
SET INPUT $PS_HOME/data/ptengs.db;  
IMPORT *;
```

Understanding Command **SWAP_BASE_LANGUAGE**

- This command swaps all the language tables from PSRECDEFN.
- Gets all table names that contain related table, swaps one table at a time.
- Copies base table into the relate table, updates related record into the base table, and then deletes related record from the related table.

SWAP_BASE_LANGUAGE - Syntax and Usage

SWAP_BASE_LANGUAGE <LANGUAGE_CD>

Example:

SWAP_BASE_LANGUAGE **FRA**;

Assumption: Your database base language is **ENG**.

Understanding Command UPDATE_DUPS

- Provided enhancement to support language data
- Operating Modes:
 - Bootstrap – update all columns
 - Regular:
 - Base table with the same LANGUAGE_CD – update all columns
 - Base table with the difference LANGUAGE_CD – update only non-translatable columns
 - Related language table - resolves and swaps the base and related language table
 - Non-language table – update all columns

UPDATE_DUPS - Syntax and Usage

```
SET UPDATE_DUPS;  
IMPORT {record | *};
```

UPDATE_DUPS – Design Pattern

DAT - ENG

Base Table					
K1	C1	Cn	L1	L2	L3
X1	X	X	ENG	ENG	ENG
X2	X	X	ENG	ENG	ENG

Related Language Table			
K1	L1	L2	L3
X1	FRA	FRA	FRA
X2	FRA	FRA	FRA

DB - FRA

Base Table					
K1	C1	Cn	L1	L2	L3
X1	⌘	⌘	FRA	FRA	FRA
X1	X	X	FRA	FRA	FRA
X2	X	X	ENG	ENG	ENG
X2	⌘	⌘	FRA	FRA	FRA

Update

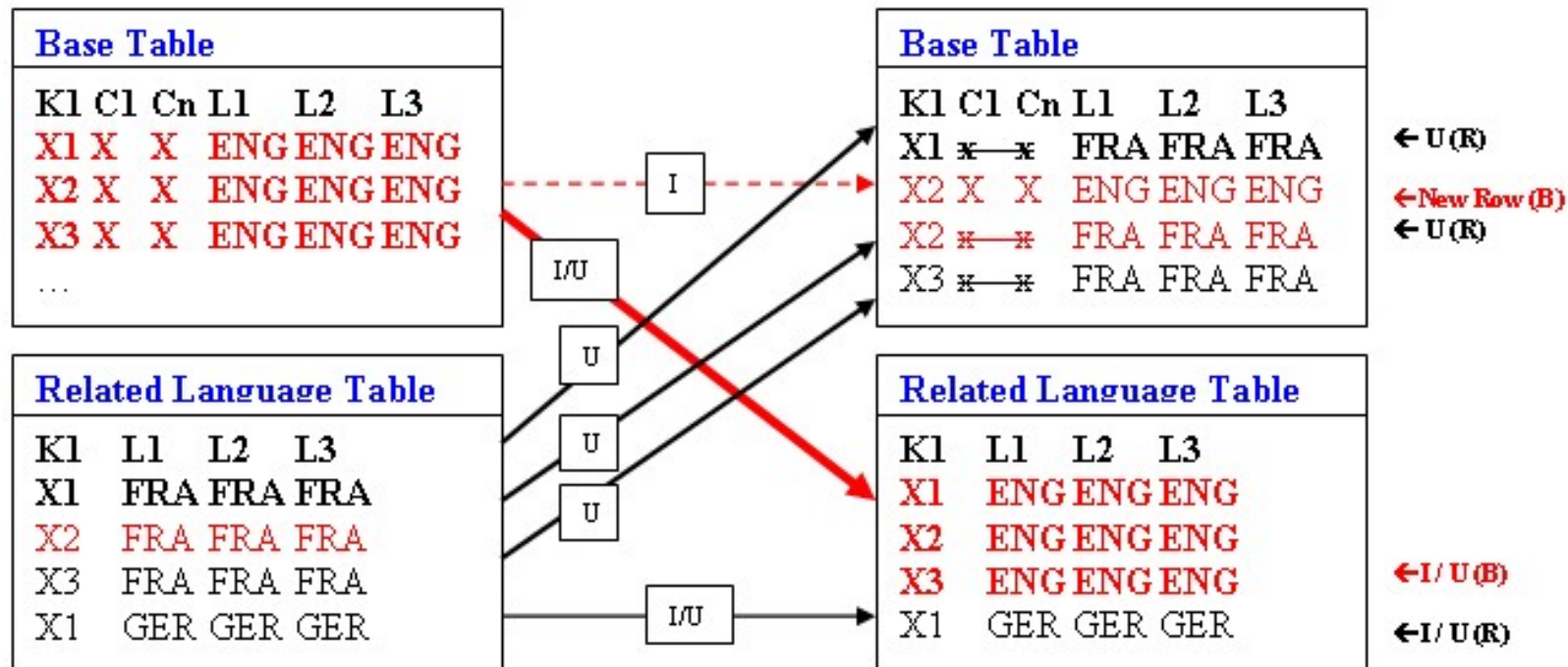
Insert

Update

UPDATE_DUPS – Design Pattern (cont.)

DAT (ENG)

Database (FRA)



Legend: I – Insert; U – Update; B – Base Table; R – Related Table; K1 – Key; C1-Cn – non-translatable; L1-L3 – Language columns

Others Commands

- <http://adntas37/htmlidoc/pt84/eng/psbooks/tdam/book.htm>

End of Session 1

Q&A

Session 2

DataMover Source Code

Data Mover

- PS_HOME/src/psdmt
- PS_HOME/src/psdmtx
- PS_HOME/src/psdam
- PS_HOME/src/psbld

DCW

- PS_HOME/src/InstallShield/PsMpDbInstall

Dependency Modules:

- pssys, pssam, psmgr, psbld, and etc

PSDMT - Source

Data Mover

- PS_HOME/src/psdmt
- PS_HOME/src/psdam
- PS_HOME/src/psbld
- PS_HOME/src/pssys
- PS_HOME/src/psmgr

PSDMTX – Source

Data Mover

- PS_HOME/src/psdmtx
- PS_HOME/src/psdam
- PS_HOME/src/psbld
- PS_HOME/src/pssys
- PS_HOME/src/psmgr

Source Code Walkthroughs

Debugging DataMover

Windows:

- Microsoft Visual Studio .NET 2003 (attach process (psdmt.exe or psdmtx.exe))
- Run SQL Trace (pscfg.exe)

Unix:

- TotalView (<http://www.etnus.com>)
totalview -a <-CT>
- Setup SQL Trace on UNIX

Debugging DCW

- Eclipse – Open Source IDE
(<http://www.eclipse.org>)
- JDeveloper – Oracle

Q&A

Session 3

New Datatype Support

- Support only for Oracle and SQLSERVER 2005
- Support only PeopleTools 8.48 and App 9 or later release

Oracle: NLS_SESSION_PARAMETERS

- SELECT VALUE FROM
NLS_SESSION_PARAMETERS WHERE
PARAMETER='NLS_LENGTH_SEMANTICS'

if <VALUE> equals 'CHAR'

Enable new Datatypes

SQLSERVER 2005: ProductVersion

- `SELECT CONVERT(NVARCHAR(1),
SERVERPROPERTY ('ProductVersion'))`

if <VALUE> equals <9 or later>
 Enable new Datatypes

Setup Data Mover on z/OS Unix System Services

- Install PeopleTools CD-ROM to File Server
- PeopleTools shell PSCONFIG.SH on UNIX
- PSMV.SH
- Environment Variables
- Data Mover runtime
- Data Mover configuration file
- ODBC initialization file

Setup Database Connectivity - ODBC Initialization File

- ODBC.ini:
 - [COMMON]
 - MVSDEFAULTSSID=DSNT
 - APPLTRACE=0
 - APPLTRACEFILENAME=/u/data004/pt845/odbctrace.log
 - MULTICONTEXT=1
 - CONNECTTYPE=1
 - CURRENTAPPENSCH=EBCDIC
-
- ; Set up the DB2 Subsystem Definition
 - [DSNT]
 - MVSATTACHTYPE=CAF
 - PLANNAME=DSNACLI

Set Configuration File

- `export PS_SERVER_CFG =
$PS_HOME/setup/psdmtx.cfg`

Environment Variables

- `export PS_DM_DATA=$PS_HOME/data`
- `export PS_DM_LOG=$PS_HOME/log`
- `export PS_DM_SCRIPT=$PS_HOME/scripts`
- `export DSNAOINI=$PS_HOME/setup/odbc.ini`
- `export PS_SERVER_CFG
=$PS_HOME/setup/psdmtx.cfg`
- `export PS_SERVDIR =$PS_HOME/log`

Data Mover Command Line

Questions

- How the data is exported to .db file (export *)
- How the data is import from .db file (import *)
- Command IGNORE_ERRORS
- Command SWAP_BASE_LANGUAGE
- Command CREATE_TRIGGER
- Command REPLACE_VIEW
- Command on DB2 zOS
- Instruction to use Data Mover on DB2 zOS
- Visual Compare & Merge